



Design considerations of calibration DAC in self-calibrated SAR A/D converters



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ABSTRACT

A capacitive calibration digital-to-analog converter (CDAC) is commonly used to reduce the mismatch-induced linearity errors for successive approximation register (SAR) analog-to-digital converters (ADC) employing capacitor arrays. There are complicated design considerations in determining the number of bits, the unit capacitor value and even the parasitic capacitors of the CDAC, as these factors affect or are determined by the achievable ADC resolution, the main DAC's capacitance, and the main DAC unit capacitance value, etc. This paper is the first to present a systematic analysis on these relationships. The analysis is validated by behavioral and circuit simulation results.

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1. Introduction

The successive approximation register (SAR) analog-to-digital converter (ADC) is widely considered the most energy-efficient type of ADCs for medium-speed applications due to its simple structure compared to other kinds of ADCs such as flash, two-stage, folded and pipelined ADC [1,2]. A SAR ADC consists of an n -bit main capacitor DAC (M-DAC) array, a comparator and a logic controller. Typically, the M-DAC is implemented by a binary weighted capacitor array, which occupies most of the chip area. The M-DAC can also be realized by a split-DAC to reduce the chip area and the ADC's input capacitance [3]. The split-DAC is a segmented binary weighted capacitor array, which are connected by a bridge capacitor [3]. For instance, two 7-bit binary weighted capacitor arrays are combined into a 14-bit binary weighted capacitor array, where the area of DAC is saved by 1/64.

The resolution of a capacitive SAR ADC is fundamentally limited by capacitor mismatches. The mismatches can be stored and aligned using a memory as in [4]. One drawback of the approach in [4] is that it requires resistor strings, which consumes static current. The capacitor mismatches can also be trimmed away with extra fabrication steps as in Ohnhaeuser [5]. In [6,7], the mismatches are tolerated by applying non-binary weighted capacitors [such as weight=1.86 instead of 2], which are actually harder to achieve good matching than binary weighted ones, to generate redundancy for correction in post processing steps.

Recently, the most straightforward self-calibration technique based on the method in [4] has been demonstrated in [8] by utilizing a Calibration capacitive digital-to-analog converter (CDAC). The 1 V supplied SAR ADC in [4] achieves a 10-bit resolution using a split-DAC with the total input capacitance of only 530 fF. The errors from parasitic capacitor and the capacitor mismatches are calibrated by the CDAC. During the calibration, the capacitor mismatches are measured by CDACs and the corresponding correction codes are stored in a memory. In the normal A/D conversion, the CDAC compensates the mismatch. This technique offers the advantages of no static current, no extra fabrication processing, no redundancy codes, and that calibration can be done in the power-up time of the whole system.

However, the addition of the CDAC complicates the relationships among design parameters. The following questions need to be addressed: how accurate the calibration can be? How to determine the resolution of the CDAC? How to determine the minimum capacitor for the M-DAC in a given fabrication process? How to size the capacitors for the CDAC? Will the parasitic capacitors of the CDAC affect the performance? Intuitively, the more bits the CDAC has, the more accuracy the calibration can achieve. The larger the M-DAC capacitors are, the better the matching is, but the area and power consumption also rise. Lots of iterations can be involved for a design to converge into a satisfactory one because those performance parameters depend and affect one another. To simplify these relationships, we first perform a systematic analysis on these design parameters, and then build a mathematical model. A high-resolution 14-bit SAR ADC employing split-DAC is designed to show the effectiveness of the model.

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The rest of this paper is organized as follows. [Section 2](#) describes the mismatch calibration process and builds the model linking different design parameters. [Section 3](#) shows the 14-bit design example. [Section 4](#) presents the simulation results and conclusions.

2. Calibration of capacitor mismatch

2.1. Mismatch induced error voltage

The random mismatch between two “equal” capacitors can be measured in two clock cycles. Fig. 1(a) illustrates the method using two signal capacitors, each representing a combination of capacitors. First, the bottom plates of C_{C1} and C_{C2} are connected to V_{ref} and ground (GND) respectively, where V_{ref} is a reference DC voltage. Next, the voltages for the bottom plates of C_{C1} and C_{C2} are swapped and the shorted top plates are disconnected from V_{cm} , as shown in Fig. 1(b). Charge redistribution occurs. If C_{C2} has a value identical to that of C_{C1} , the top plate voltage remains at V_{cm} . However, a mismatch between C_{C1} and C_{C2} causes a small voltage change ΔV_x :

$$\Delta V_x = \frac{\Delta C}{2C} V_{ref} \quad (1)$$

where $\Delta C = C_{C2} - C_{C1}$, $C = (C_{C1} + C_{C2})/2$. The mismatch induced error voltage ΔV_x is thus a direct indication of the capacitor mismatch [4]. It is also referred to as the residual error voltage ΔV_x in this paper.

To correct the mismatch, a CDAC is added to the top-plate as shown in Fig. 2, where C_p and C_{p_cal} represent the total parasitic capacitances at the corresponding nodes. C_p includes comparator's input capacitance, parasitic capacitance of the M-DAC top plate, and bottom plate of the C_B ; C_{p_cal} includes the drain-to-gate,

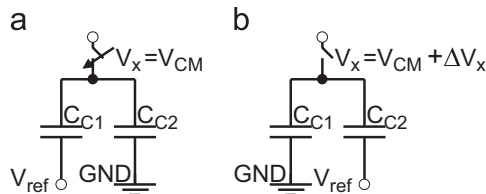


Fig. 1. (a) Pre-charging and (b) residual error voltage detection.

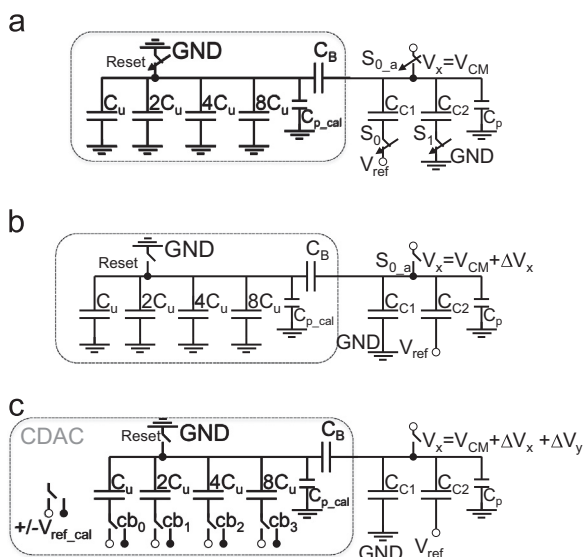


Fig. 2. Three phases in the calibration (e.g.: 4-bit CDAC): (a) pre-charging of the capacitors, (b) ΔV_x detection, and (c) calibration.

drain-to-bulk capacitances, parasitic capacitances of the top plates of CDAC and C_B . Instead of using a resistor string [9], a binary weighted capacitive DAC is generally preferred to implement the CDAC. Fig. 2 shows a 4-bit CDAC example. The bridge capacitor C_B serves the purpose of reducing the effective CDAC capacitance when seen at the top plate of the M-DAC capacitor array. Note that the added capacitance at the top plate of the M-DAC affects only the gain of the M-DAC, not its linearity, so the bottom plate (which has larger parasitic than the top plate) of C_B is connected to the M-DAC. We denote the capacitor values of CDAC in the presence of mismatch as:

$$C_{cali} = 2^i C_u (1 + \epsilon_{cali}), i = 0, 1, \dots, m-1 \quad (2)$$

$$C_{\text{caltot}} = C_B(1 + \varepsilon_B) + \sum_{i=0}^{m-1} 2^i C_u(1 + \varepsilon_{\text{cali}}) = 2^m C_u \quad (3)$$

where C_{cali} is the i th capacitor in the CDAC with a random percentage error ε_{cali} from its nominal value, and C_{caltot} is the total CDAC capacitance which by definition has no error. The bridge capacitor C_B is assumed to be of the same value as the unit capacitor of the CDAC, C_u . This makes the layout easy and the matching being the best. C_B has a random percentage error ε_B from its nominal value.

Fig. 3 shows the timing diagram of the calibration process. The CDAC is first reset when the circuit enters the calibration mode. In order to avoid the switch charge injection to the CDAC, we leave the top plate of M-DAC floating. The charge injection from switch S_{0a} contributes to ΔV_x , which can be reduced using complementary switch and differential structure. The pre-charging is done by connecting the top plate of C_{C1} and C_{C2} to a DC voltage V_{cm} , the bottom plates of C_{C1} and C_{C2} to V_{ref} and ground respectively, as shown in Fig. 2(a). The charge is then redistributed in the common top plate when it is disconnected from V_{cm} and the bottom-plate voltages of C_{C1} and C_{C2} are swapped, as shown Fig. 2(b). During these two phases, the CDAC is always grounded (see Fig. 2).

The residual error voltage ΔV_x at X is renewed as:

$$\Delta V_x = \frac{\Delta C}{C_T} V_{ref} \quad (4)$$

where

$$C_T = C_{C1} + C_{C2} + C_p + k \times C_B(1 + \varepsilon_B)(1 - \beta_A) \quad (5)$$

$$\beta_A = \frac{C_B(1 + \varepsilon_B)}{C_{caltot} + C_{p\ cal}} \quad (6)$$

where k ($k=1,2,3,\dots$) is the number of CDACs that are connected to the top plate of C_{C1} and C_{C2} in parallel, and ε_B is the percentage error of bridge capacitor C_B . It is seen from (4) that the error voltage ΔV_x is proportional to the mismatch ΔC while C_T is a total capacitance associated with the top plate of M-DAC. The C_p and CDAC array only reduce the magnitude of ΔV_x , but does not change the sign, as the other terminals of these capacitors are tied to fixed potentials.

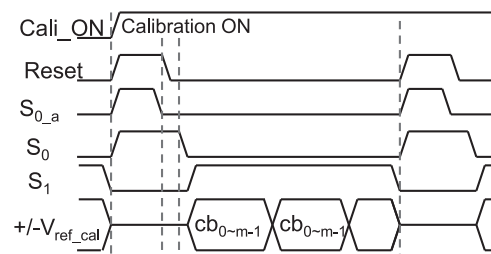


Fig. 3. Calibration timing diagram.

2.2. Mismatch calibration

Subsequently, a compensation step can be performed by introducing an additional voltage ΔV_y from CDAC to the residual error voltage ΔV_x . Fig. 2(c) illustrates the process. A proper code for the CDAC needs to be determined with the goal of minimizing ΔV_x in (4). The correction is done by switching the bottom plates of the CDAC capacitors from 0 to reference voltage $+V_{ref_cal}$ or $-V_{ref_cal}$ in a successive approximation manner. It starts from the most-significant bit (MSB) to the least significant bit (LSB) of the CDAC and stops when ΔV_x becomes smaller than a target value ($1/2LSB$). As C_{C2} can be larger or smaller than C_{C1} , ΔV_x can be positive or negative. For correcting the mismatch in both cases, the CDAC needs both positive and negative V_{ref_cal} . The V_{ref_cal} can be or be not the same as the reference V_{ref} of C_{C1} and C_{C2} . When the calibration is finished, the CDAC will be switched in synchronization with C_{C2} in the normal operation of SAR ADC Fig. 4.

After adding the CDAC with bit setting of cb_i ($=0$ or 1), $i=0,1,\dots,m-1$, the final residual error voltage ΔV_{xf} at the top plate of M-DAC becomes:

$$\Delta V_{xf} = \frac{\Delta C}{C_T} V_{ref} + \Delta V_y \quad (7)$$

where

$$\Delta V_y = K_y \sum_{i=0}^{m-1} C_{cali} (cb_i - \overline{cb_i}) V_{ref_cal} \quad (8)$$

$$K_y = \frac{\beta_B}{C_{caltot} + C_{p_cal} - C_B(1 + \epsilon_B)} \quad (9)$$

$$\beta_B = \frac{C_B(1 + \epsilon_B)}{C_T + \beta_A C_B(1 + \epsilon_B)} \quad (10)$$

therefore, the CDAC can in principle be adjusted to reduce ΔV_x , caused by capacitor mismatch, to 0. It's observable from (7) and (8) there is a maximum correctable mismatch ΔC due to the limited correction range of the CDAC. The absolute maximum mismatch $|\Delta C|_{max}$ subsequently corresponds to the calibration

code of 00...00 or 11...11. This leads to:

$$|\Delta C|_{max} = K_{cal} \frac{V_{ref_cal}}{V_{ref}} \quad (11)$$

where,

$$K_{cal} = \beta_A [C_{caltot} - C_B(1 + \epsilon_B)] \quad (12)$$

when there is no mismatch, the residual error voltage ΔV_x is zero and the CDAC control code should reside at its mid-code. In the 4-bit CDAC example, the CDAC control code should be 1000. It also leads to a residual mismatch voltage V_{rm} after calibration, which must be less than at least half of the ADC's LSB.

$$V_{rm} = \beta_A C_u (1 + \epsilon_u) V_{ref_cal} < 1/2LSB \quad (13)$$

2.3. M-DAC calibration

The n -bit main capacitor DAC (M-DAC) can be implemented in any one of these three different architectures: (a) a conventional binary weighted capacitor array, (b) split-DAC with a non-integer bridge capacitor [3], and (c) split-DAC with an integer bridge capacitor [10]. The calibration technique can be applied in all the cases to achieve high resolution. Without loss of generality, the illustration below focuses on the binary weighted capacitor array case.

With the presence of mismatch, the value of i th capacitor, C_i , is expressed as:

$$C_i = 2^i C_0 (1 + \epsilon_i), \quad i = 0, 1, \dots, n-1 \quad (14)$$

where n is the number of bits of the M-DAC, C_0 is the nominal (averaged) value of all the unit capacitors forming the capacitor array and ϵ_i represent the percentage variation of C_i . Besides, there is a dummy capacitor C_A of the value of unit capacitor. The total capacitance C_{M-DAC} is

$$C_{M-DAC} = C_A + \sum_{i=0}^{n-1} 2^i C_0 (1 + \epsilon_i) = 2^n C_0 \quad (15)$$

The last equality holds in (6) by definition (C_0 is the mean unit capacitance, i.e., $C_{M-DAC}/2^n$).

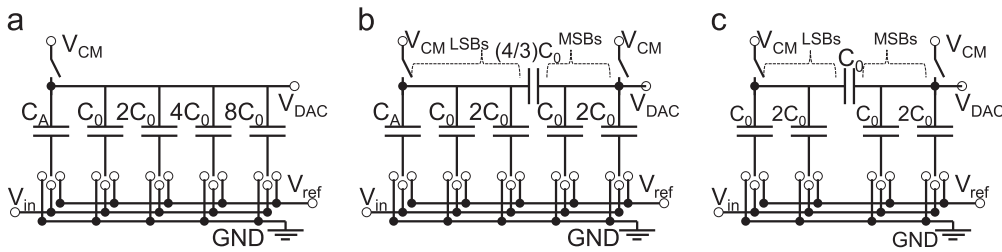


Fig. 4. A 4-bit M-DAC using: (a) a binary-weighted capacitor array; (b) split-DAC with a fractional-valued bridge Cap.; (c) split-DAC with a unit bridge Cap.

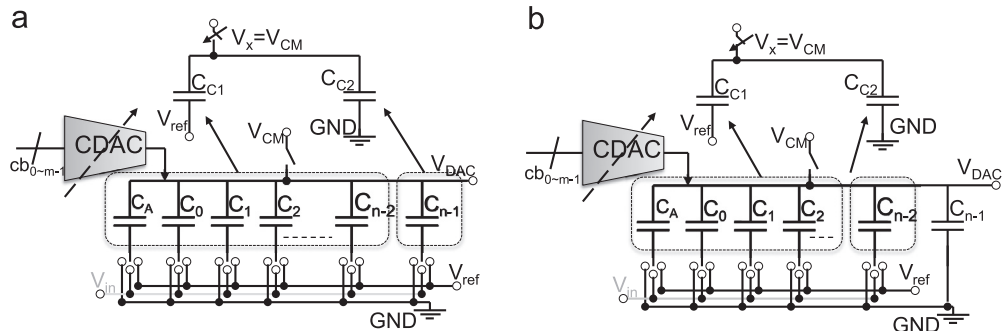


Fig. 5. Calibration for (a) the MSB capacitor and (b) the 2nd MSB capacitor.

Using the method presented in Section 2.2, the mismatch between two “equal” capacitors can be calibrated out. Fig. 5(a) illustrates the calibration of the MSB capacitor, in which C_{C1} represents the sum of all capacitors except MSB capacitor and C_{C2} presents the MSB capacitor. Ideally, $C_{C1} = C_{C2} = 2^{(n-1)}C_0$. If the 2nd MSB capacitor is calibrated as Fig. 5(b) shows, the MSB capacitor is tied to ground during the calibration, and C_{C1} and C_{C2} are selected to the value of $2^{(n-2)}C_0$. Similarly, the methodology can be applied to calibrate the rest of the capacitors in the M-DAC. Take the NSth ($NS \in 0, 1, \dots, n-1$) capacitor C_{NS} as an example. The value of C_{NS} is matched to the sum ($C_A + \sum_{i=0}^{NS-1} C_i$). Then we repeat this for C_{NS+1} , ..., and C_n . Note that each capacitor under calibration needs an independent CDAC. During the calibration of the selected capacitor, only the corresponding CDAC functions. Capacitors in all the other CDACs are grounded, behaving merely as parasitic capacitors at the top plate of the M-DAC. Thus, the calibration retains valid for every capacitor to be calibrated, and the sequence of calibration (from MSB to LSB or from LSB to MSB) becomes irrelevant.

3. Circuit implementation

3.1. ADC architecture

A 14-bit SAR ADC is designed to validate the calibration methodology. Fig. 6 shows the architecture of the ADC. It is a differential circuit consisting of two M-DACs, a number of CDACs, an offset calibrated comparator, a shift register, and a logic controller for the CDACs. The offset of comparator affects the accuracy of the calibration and thus the linearity of the calibrated ADC. The offset of comparator have to be compensated at least less than half of the LSB. Here we assume that the comparator offset is suppressed to be far less than the final resolution by techniques like the body-biasing techniques [4] or others [11–14]. Besides of the static error, comparator noise also requires to be within 0.5LSB as it directly influences the calibration accuracy. Fortunately, the noise estimation in strobed comparator has been well understood in [13,15]. Followed by the design guidelines in [13], the comparator noise should be fitted within our target (50 μ V).

The operation sequence of ADC is as follows. The first step is the calibration of the input offset of the comparator. As the second step, the control logics ($cb_0 \sim cb_{m-1}$)_k of CDACs are estimated and the estimated control logics are stored in a memory. After that, the normal A/D conversion starts with the CDACs capacitors switching in synchronization with their corresponding M-DAC capacitors.

3.2. Design parameters considerations

Choosing the appropriate CDAC to fit the desired resolution of an M-DAC is a challenging task as there are too many parameters affecting one another. The mathematical model built in Section 2 gives an intuition look into the design. An example of 14-bit resolution M-DAC using a half split-DAC with integer bridge

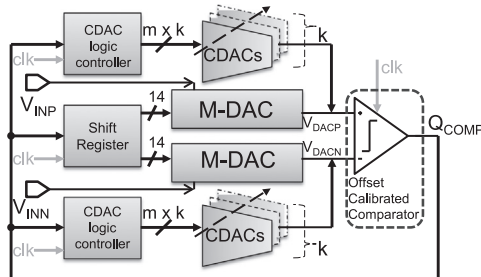


Fig. 6. Architecture of differential SAR-ADC using CDACs.

capacitor shows the design procedures. First of all, we know the resolution and architecture of the M-DAC from the system specifications. Second, we choose C_u to be 17.2 fF based on the thermal noise requirement ($\sqrt{kT/C_s} < 1/4$ LSB, where C_s is the sampling capacitor). The 17.2 fF capacitor is an MIM capacitor with the size of $4 \mu\text{m} \times 4 \mu\text{m}$. After that, we follow the mathematical analysis step by step from Sections 3.2.1–3.2.6.

3.2.1. Residual error voltage (ΔV_x) vs. capacitor 3-sigma mismatch

The capacitor mismatch is generally considered as a random process following the Gaussian distribution with zero mean and certain standard deviation. Therefore, we define:

$$\delta = 3\sigma_0 \quad (16)$$

where σ_0 is the standard deviation of the mismatch between two unit capacitors. So the chance that the mismatch between two unit capacitors being smaller than δ is 99.7%. In the M-DAC, the capacitor C_i is composed of 2^i unit capacitors. The variance of the capacitor C_i is thus $2^i \sigma_0^2 C_0^2$. When normalized to the nominal value $(2^i C_0)^2$, the variance becomes $2^{-i} \sigma_0^2$. In other words, the percentage error ϵ_i in (6) has a standard deviation of

$$\sigma(\epsilon_i) = 2^{-i/2} \sigma_0 \quad \text{or: } 3\sigma_i = 2^{-i/2} \delta, \quad i = 0, 1, \dots, n-1. \quad (17)$$

with these 3-sigma values used as the mismatch values, the worst case is covered and the time-consuming Monte Carlo simulation is unnecessary [16,17]. Therefore, the mismatch ΔC of the two capacitors under calibration has the maximum derivation of $2^{NS/2} C_0 \delta$ ($NS \in 0, 1, \dots, n/2-1$ for the half split-DAC). Replacing this mismatch into (4), the residual error voltage ΔV_x is obtained as:

$$\Delta V_x = \frac{2^{NS/2} C_0 \delta}{C_T} V_{ref} \quad (18)$$

then, the residual error voltage ΔV_x for the smallest ($NS=0$) and largest capacitor ($NS=6$) to be calibrated in the MSBs of the half split-DAC is drawn in Fig. 7. The effects on the parasitic capacitor at the top plate of MSBs of the half split-DAC do not alter the sign of the residual error voltage, thus is not problematic. If the unit capacitors have 3-sigma mismatch of 2.5%, the mismatch from the largest capacitor contributes approximately 2.7 mV residual error voltage, while that of the smallest capacitors is about 0.3 mV. From (18), ΔV_x increases exponentially with NS as the coefficient C_T is a constant for a given M-DAC.

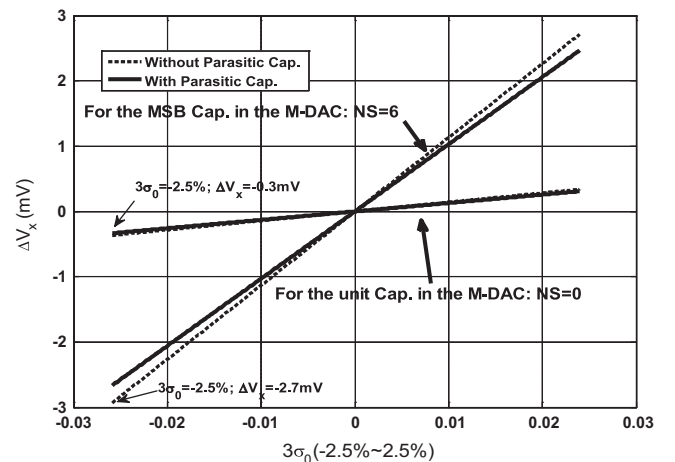


Fig. 7. Residual error voltage (ΔV_x) vs. unit capacitor $3\sigma_0$ mismatch.

3.2.2. Bridge capacitor (C_B) and unit capacitor (C_u) of CDAC

Assume $V_{ref_cal} = V_{ref}$. From (11) and (12), $|\Delta C|_{max}$ is approximately equals to C_B only if the total capacitor C_{caltot} at the CDAC is greater than C_B . Meanwhile, the residual mismatch voltage V_{rm} after calibration is also affected by C_B in (13). Larger C_B leads to larger V_{rm} . Therefore, a proper value of C_B can be chosen according to the requirements on both $|\Delta C|_{max}$ and V_{rm} . In such case, the unit capacitor C_u of CDAC only needs to meet the matching requirement as the capacitor divider relaxes the impact. However, for better layout and design simplicity, we set $C_B = C_u$. This results in that the value of C_u should be chosen in an appropriate range. The standard deviation σ_0 of the mismatch for the 17.2 fF unit capacitor can be obtained from the foundry. We consider $3\sigma_0 = 2\%$ with a very relaxed matching requirement. With the matching property, the $|\Delta C|_{max}$ determines the minimum value of C_u . On the other hand, V_{rm} determines the maximum boundary of C_u . Fig. 8 shows the relationship between C_u and the final residue error ΔV_{xf} for certain mismatch ranges. The solid line without mismatch shows the maximum C_u is at 6.5 fF. The largest capacitor to be calibrated in the M-DAC (NS=6) shows the lowest boundary of C_u at 2.5 fF. Such appropriate range of C_u is easily realizable in an advanced CMOS technology. In this plot, the number of bits of CDAC is set to 6.

3.2.3. CDAC bits (m)

As the step size of CDAC affecting the calibration resolution directly, high resolution of CDAC increases the accuracy of the calibration. The analytic plots in Fig. 9 implies the bits of CDAC is required to be greater than 5 for the 50 μV resolution, which is a

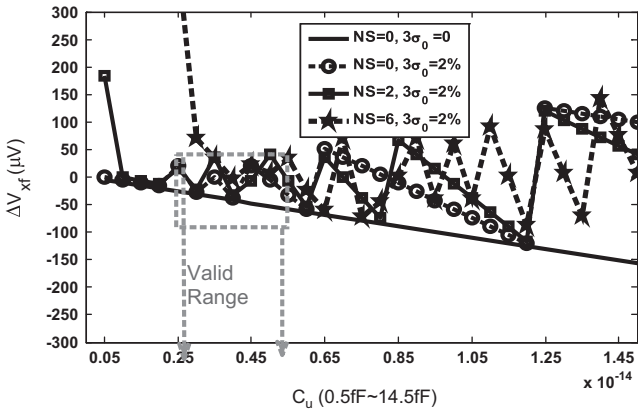


Fig. 8. Final residual error voltage (ΔV_{xf}) vs. unit capacitor (C_u) of CDAC.

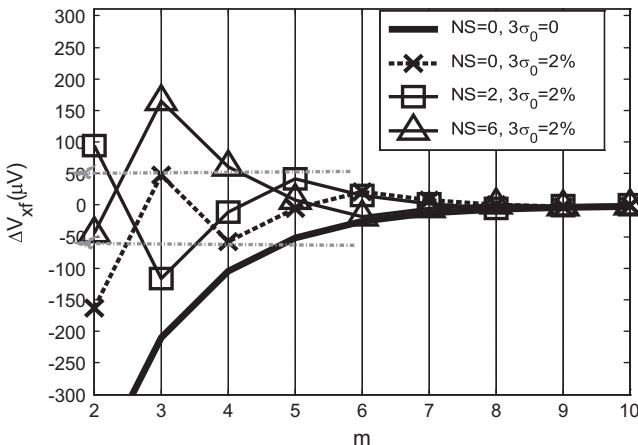


Fig. 9. Final residual error voltage (ΔV_{xf}) vs. CDAC bits (m).

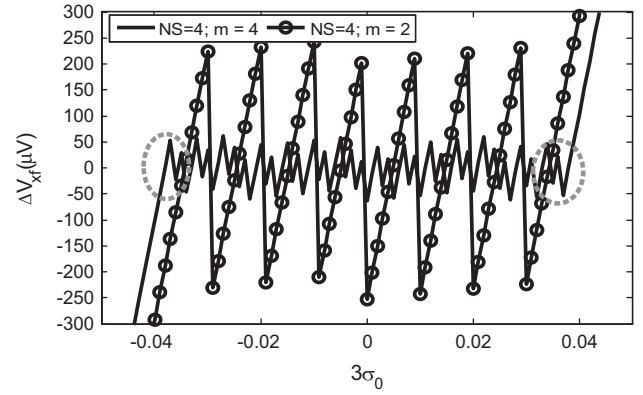


Fig. 10. CDAC bits (m) vs. unit capacitor $3\sigma_0$ mismatch.

half of the LSB in 14-bit resolution of our design target. However, the increasing of CDAC bits only reduces ΔV_{xf} but not enlarges the $|\Delta C|_{max}$ (see (11)). Simulation results confirming this are shown in Fig. 10. For the same capacitor to be calibrated (NS=4) and the same C_u , the final residual error voltages have more reduction in the case of 4-bit CDAC than that of 2-bit CDAC, but the range of $3\sigma_0$ remains the same. In other words, a higher than necessary number of bits in the CDAC costs unnecessary chip area and power. Therefore, we set the CDAC bits to be 6.

3.2.4. Calibrated capacitor numbers (NS) and MSB bits in the split-DAC

In previous discussion, larger calibrated capacitor number (NS) causes larger residual error voltage ΔV_x . However, a given CDAC can only cover a limited range of ΔV_x . This implies that there is a maximum NS for a given M-DAC and $|\Delta C|_{max}$. Fig. 11(a) shows that the final residual error voltage ΔV_{xf} grows fast when NS is greater than 6 with a $3\sigma_0$ of 2%.

The ΔV_{xf} needs to be calibrated into an accuracy better than $1/2$ LSB, in which the LSB is defined as ratio of the unit capacitor weight to the total capacitance in the M-DAC ($LSB = V_{ref}/2^n$ if we ignore the parasitic capacitors). LSB is a constant for a given M-DAC so that reducing ΔV_{xf} improves the calibration accuracy. Therefore, the choice of MSB bits in split-DAC affecting the value of C_T matters the overall calibration accuracy. Fig. 11(b) shows that a minimum choice of 7-bit for the MSB bits in split-DAC guarantees the 50 μV accuracy when the capacitor mismatch $3\sigma_0$ reaches about 2%. Extremely, it becomes a conventional binary-weighted capacitor array (BWA) when MSB bits reach 14. The smallest ΔV_{xf} also implies the accuracy would be better in BWA than those in split-DAC.

3.2.5. Reference voltage (V_{ref_cal}) and parasitic capacitor (C_{p_cal}) of CDAC

The V_{ref_cal} affects $|\Delta C|_{max}$ and V_{rm} directly from (7) and (11). It has the same mechanism as the C_u , which cannot either too large or too small. Such appropriate range of V_{ref_cal} is plotted in Fig. 12(a). It tells the final residual error voltage ΔV_{xf} is smaller than 50 μV when V_{ref_cal} is chosen to be the same as V_{ref} , which is 1.8 V in this design.

The parasitic capacitor C_{p_cal} at the top plate of CDAC contributes to the term β_A in (5), which increases C_T and reduces K_y directly, and in turn decreases $|\Delta C|_{max}$ and also V_{rm} . However, it's predictable that the parasitic capacitor C_{p_cal} can be ignored because it is much smaller than C_{caltot} in (6). Estimating the tolerance of C_{p_cal} in Fig. 12(b) also shows that ΔV_{xf} is satisfied until the C_{p_cal} is greater than 100 fF with the capacitor $3\sigma_0$ mismatch of 1%.

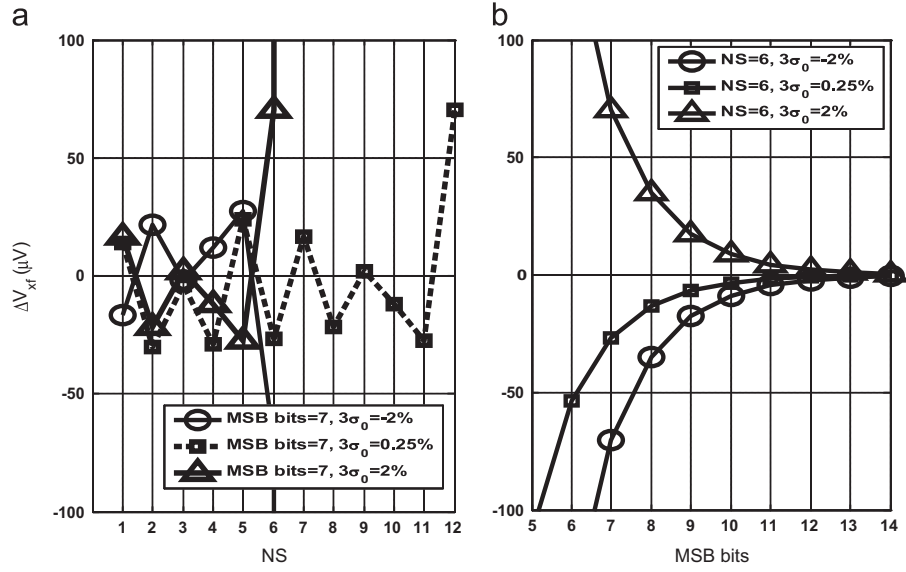


Fig. 11. Final residual error voltage ΔV_{xf} vs. (a) calibrated capacitor number (NS); (b) MSB bits in the split-DAC.

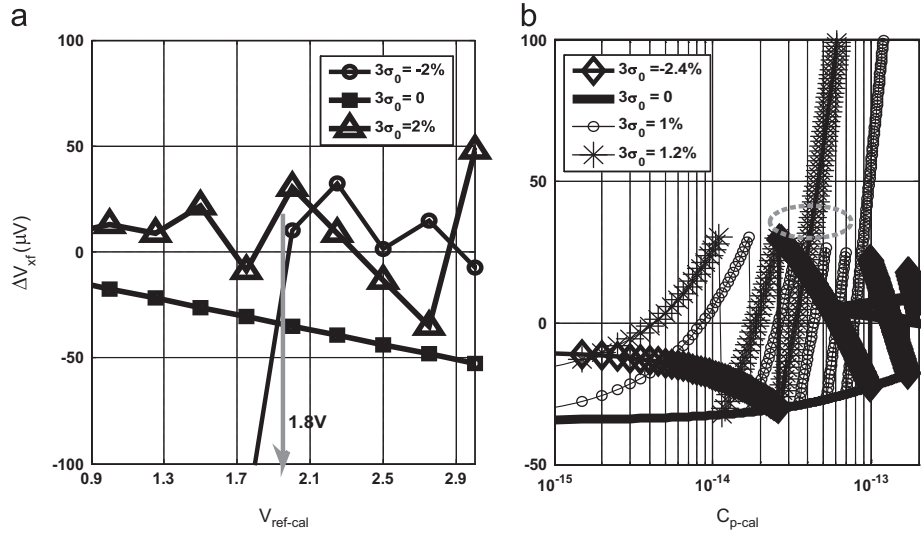


Fig. 12. Final residual error voltage ΔV_{xf} vs. (a) CDAC's reference voltage ($V_{ref-cal}$); (b) parasitic capacitor at the top of CDAC (C_{p-cal}).

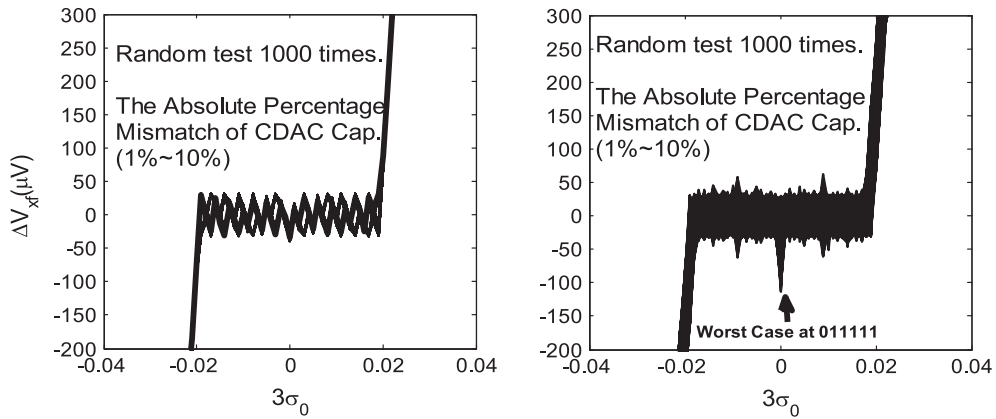


Fig. 13. Effects on the percentage mismatch of CDAC's unit capacitor.

3.2.6. Percentage mismatch of CDAC capacitors

As rule of thumb, the mismatch for CDAC relies on the number of bits m for CDAC's monotonicity, such as: the percentage

mismatch ϵ_{cal0} with 3-sigma value of C_u in the CDAC is less than 2^{-m} [16,17]. This matching requirement is much easier to meet comparing to the M-DAC. For the 6 bits CDAC, the

Table 1
Parameters for the design.

Power supply	1.8 V	M-DAC's reference voltage (V_{ref})	1.8 V
Half split-DAC (M-DAC)	7 bit + 7 bit	Resolution required (n)	14 bit
M-DAC's unit capacitor C_0	17.2 fF	CDAC's unit capacitor C_u	2.5–6.5 fF
Total capacitance of M-DAC	2.2016 pF	Total capacitance of CDAC	192 fF
CDAC bits (m)	6 bit	Selected Capacitor to be calibrated in M-DAC (NS)	1–6
Number of CDACs (k)	6	3σ range for C_u	$\pm 2.5\%$
3σ range for C_0	$\pm 1.8\%$ (@ NS=6)	Parasitic capacitance at the top of CDAC (C_{p_cal})	0–100 fF
CDAC's reference voltage (V_{ref_cal})	± 1.8 V	Common mode voltage (V_{cm})	0.9 V

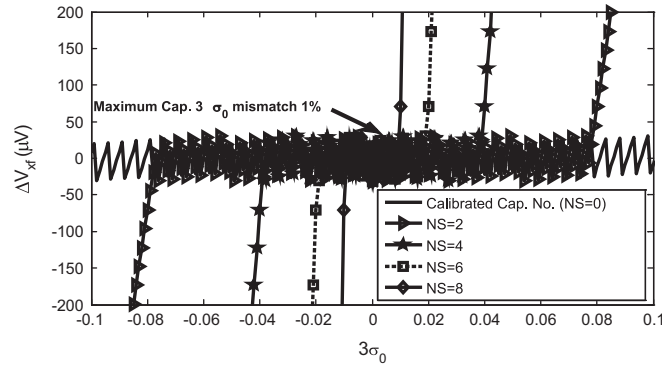


Fig. 14. Final residual error voltage ΔV_{xf} vs. unit capacitor $3\sigma_0$ mismatch.

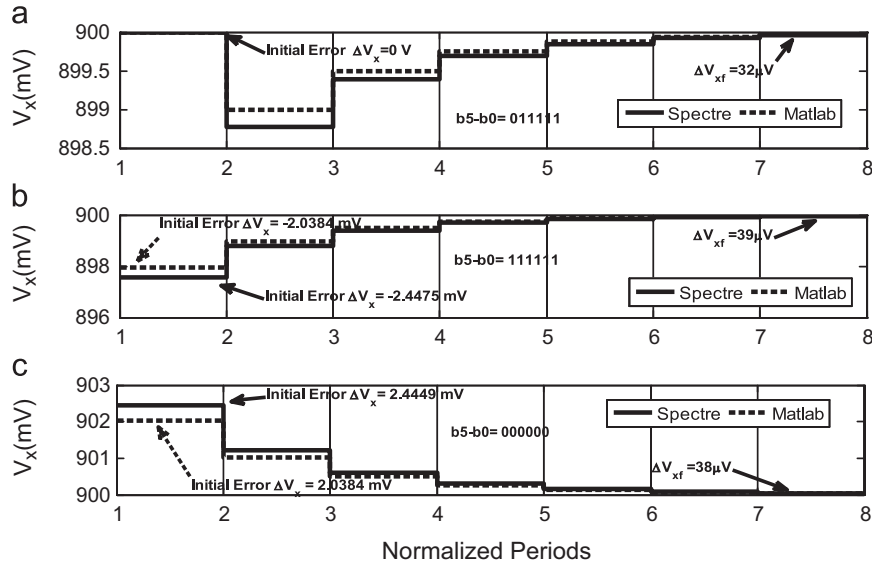


Fig. 15. Simulated waveforms of residual voltage V_x : (a) with zero initial error; (b) and (c) with absolute maximum initial error.

percentage mismatch of CDAC capacitor should be less than 1.7% theoretically.

The simulation condition is with the worst case in the design, and we use split-DAC in 14-bit configuration with parasitic capacitors and NS=6 for the largest capacitor to be calibrated. Fig. 13 shows that for $\pm 2.5\%$ M-DAC capacitor 3σ mismatch, the final residual error voltage ΔV_{xf} is always safely within $50 \mu\text{V}$ only when the CDAC capacitor percentage mismatch is less than 1%. In practice, 1% mismatch for 6-bit CDAC capacitor array is comfortably achievable in modern CMOS technologies (e.g.: from the $0.18 \mu\text{m}$ technology, $3\sigma_0$ is around 0.5% for 5 fF capacitor).

Furthermore, for a CDAC capacitor percentage mismatch between 1% and 10%, most of the region is located within the $50 \mu\text{V}$ except for the worst case with the calibration bits 011111. This is explainable that the worst DNL always happens at the transition from 011111 to 100000.

3.3. Parameter summary

Based on the systematical analysis above, the design parameters regarding to the M-DAC and CDAC are shown in Table 1. In this design, we utilize the half-split M-DAC with an integer

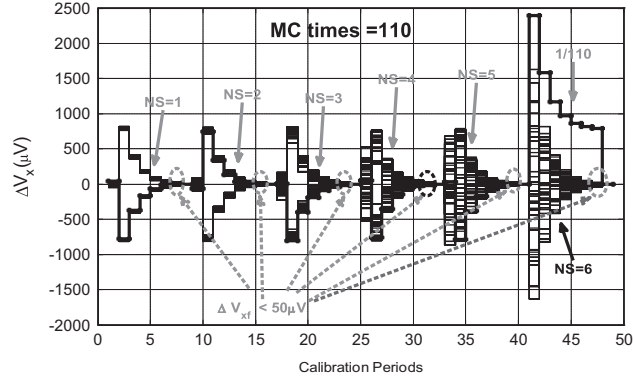


Fig. 16. Calibration Process with 110 Monte Carlo simulations.

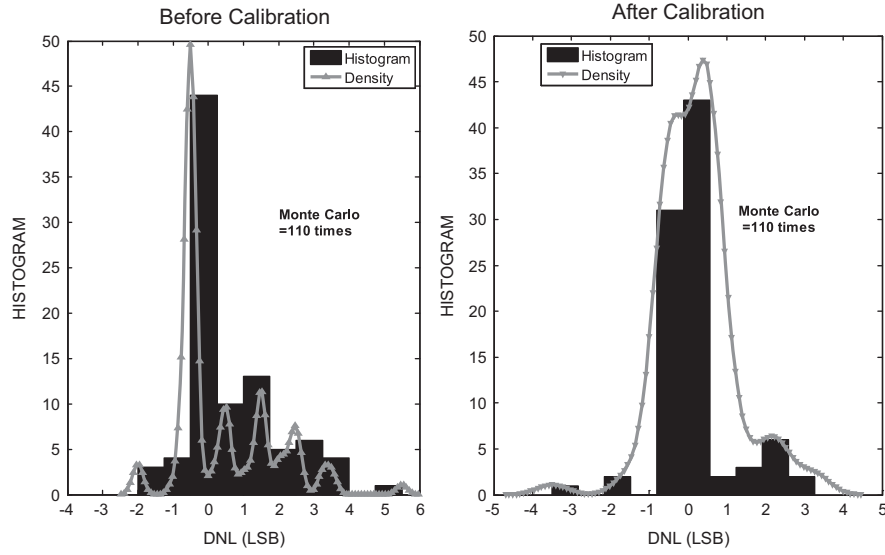


Fig. 17. DNL at digital output 0111...111_n.

bridge capacitor. The total capacitance of M-DAC is reduced to 2.2016 pF for a 14-bit resolution, which is a significant reduction from the state of the art.

4. Simulation results

4.1. Maximum $3\sigma_0$ mismatch range

With those design parameters, the final residual error voltage ΔV_{xf} versus the $3\sigma_0$ mismatch for some capacitors ($NS=0, 2, 4, 6$) to be calibrated in the M-DAC is plotted in Fig. 14. The largest mismatch δ that could be fixed is $\pm 1.8\%$ for the largest capacitor ($NS=6$) in the M-DAC, which is easily achievable in practice. In real circuit design, larger capacitors have better matching, and the calibration capability has larger tolerance. Moreover, for the smaller calibrated capacitor number NS , such as $NS=0, 2$, and 4 , the capacitor mismatching range has been relaxed. An extension case of $NS=8$ can also be tolerated to 1%.

4.2. Calibration progress

For the calibration progress shown in Fig. 15, the ΔV_{xf} can be attenuated to be less than $50 \mu V$. Fig. 15(a) is the case of no mismatch. The ΔV_{xf} brought by V_{rm} must be less than the resolution required. Fig. 15(b) and (c) are two extreme cases, which

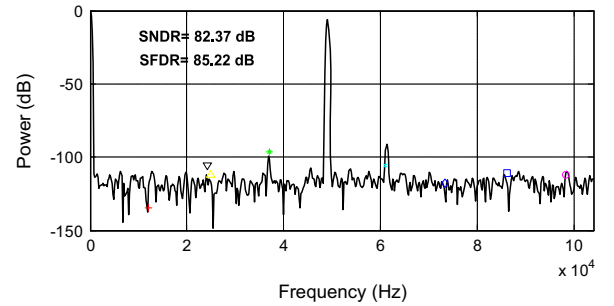


Fig. 18. ADC output spectrum at 50-kHz input.

represents the boundaries of capacitor mismatch. The corresponding CDAC code is 000000 and 111111, respectively. The MATLAB simulation closely matches the SPECTRE behavior, with the small difference attributing to the inaccuracy in the parasitic capacitor value in the MATLAB model. Monte Carlo simulations in SPECTRE show the calibration results for those capacitors to be calibrated ($NS=1, 2, \dots, 6$) in Fig. 16. No. 1, the residual error voltage ΔV_x was approximately approaching to be less than $50 \mu V$ through 6-bit CDAC searching. No. 2, only 1 out of 110 simulations failed to calibrate the mismatch into the required accuracy from a too large initial error.

Table 2
Important parameters for the calibration methodology.

Symbol definition	Equations	Explanation
ΔV_x	$\frac{\Delta C}{C_T} V_{ref}$	Residual error voltage introduced by capacitor mismatch
ΔV_{xf}	$\frac{\Delta C}{C_T} V_{ref} + \Delta V_y$	Final residual error voltage after calibration
C_T	$C_{C1} + C_{C2} + C_p + k \times C_B(1 + \varepsilon_B)(1 - \beta_A)$	Total capacitance at the top plate of M-DAC
C_{C1}	$C_A + \sum_{i=0}^{NS-1} C_i$	The selected capacitor in M-DAC to be calibrated (NS is the calibrated capacitor number.)
C_{C2}	$2^{NS} C_0(1 + \varepsilon_{NS})$	The selected capacitor in M-DAC to be calibrated
β_A		Capacitance division
C_{caltot}	$2^m C_u$	Total CDAC capacitance
ΔV_y	$K_y \sum_{i=0}^{m-1} C_{cali}(cb_i - \overline{cb_i}) V_{ref_cal}$	Correction voltage introduced by CDACs
K_y	$\frac{\beta_B}{C_{caltot} + C_{p_cal} - C_B(1 + \varepsilon_B)\beta_B}$	Coefficient of correction voltage
β_B	$\frac{C_B(1 + \varepsilon_B)}{C_{OA} + \beta_A C_B(1 + \varepsilon_B)}$	Capacitance division
$ \Delta C _{max}$	$K_{cal} \frac{V_{ref_cal}}{V_{ref}}$	Absolute maximum capacitor mismatch
K_{cal}	$\beta_A [C_{caltot} - C_B(1 + \varepsilon_B)]$	Coefficient of capacitor mismatch range
V_{rm}	$\beta_A C_u(1 + \varepsilon_u) V_{ref_cal}$	Residual mismatch voltage after calibration

4.3. Performance after calibration

Fig. 17 shows the effects of the calibration in the worst DNL code test, which happens at 0111...111_n with the input of half-reference voltage in the middle of input range because of the largest transition [17]. Post-layout simulated in a 0.18 μ m CMOS technology that achieved worst DNL with a zero mean and one LSB standard deviation after calibration in 110 times Monte Carlo simulation. The DNLs become much better than the ones before calibration. With a pure input of 50 kHz sinusoidal signal, the digital output of the ADC after calibration is taken for spectrum analysis. During the simulation, 3-sigma capacitor mismatch is accompanied. Fig. 18 illustrates the result of the dynamic performance at the sampling frequency of 200 kHz. SNDR is 82.37 dB (ENOB=13.4) and the spurious-free dynamic range (SFDR) was 85.22 dB. This performance of SNDR and SFDR (INL) show that the capacitor mismatch becomes less significant after the calibration. The third and fifth harmonic happens mainly because of the parasitic capacitor at the top plate of LSBs in the split-DAC.

5. Conclusions

This paper presents a systematic analysis on the design of calibration DACs for high resolution SAR ADCs. Low area and low power are brought by advanced CMOS technologies, but device mismatch becomes main hurdle to implement high resolution SAR ADCs. The use of calibration DACs is an effective way to overcome the hurdle. The calibration methodology has been modeled and several conclusions have been drawn. First, the resolution and unit capacitor of the CDAC should be chosen appropriately according to the M-DAC's resolution and the mismatch percentages of those binary capacitors in the M-DAC. Table 2 summarizes some important parameters related to the calibration methodology. The results are verified by SPECTRE simulations. Second, the final ADC accuracy is not very sensitive to the CDAC parasitic capacitors and the CDAC capacitor mismatches. Third, the correctable capacitor mismatch range is mainly determined by the unit capacitor of CDAC and the capacitance under calibration itself. This work has simplified the relationship among the resolution, size and parasitic capacitors of the CDAC, M-DAC resolution and the calibrated capacitor, etc, which serves as useful guideline in the design of calibrated SAR ADC.

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